

TRUSTLESS WEB3 MILESTONE PROTOCOL

Gasless Escrow for Student Freelancers

Eliminating payment ghosting and Web3 gas barriers for the next generation of builders.

TEAM & INSTITUTION

UIT — VNU-HCM Builders

VERIFIED CONTRACT

0x57099f3125faF6591f23000E8985E18c0c2202Acgithub.com/PHONGUIT22/MicroEscrow

The Harsh Reality of Student Freelancing

62%



Payment Ghosting Rate

Clients disappear immediately after receiving completed deliverables.

\$1,500+



Yearly Student Earnings Lost

Critical living allowance and tuition money vanishes to bad-faith clients.

\$0



Zero Legal Recourse

Traditional legal systems cannot enforce \$50–\$500 micro-contracts.

10–20%



Web2 Platform Middleman Tax

Upwork and Fiverr impose heavy fees and 14-day bank holding periods.

The Web3 Paradox: "I need crypto to earn crypto"



THE BLOCKCHAIN PROMISE

Mathematical Trust

Non-custodial smart contracts mathematically guarantee milestone payouts.

THE FATAL FRICTION

The Zero-Balance Trap

Students with \$0.00 / 0 ETH cannot pay gas fees just to submit proof of work.

⚠️ Result: 95% Drop-Off Rate

Talented college engineers abandon Web3 before completing their first gig.

MicroEscrow: Trustless Security with Zero Friction



Locked Milestones

Clients deposit contract funds before work starts.

No unilateral client withdrawals



Zero-Gas Proofs

Freelancers submit GitHub PRs or Figma URIs freely.

\$0.00 wallet balance requirement



Auto-Release Timeout

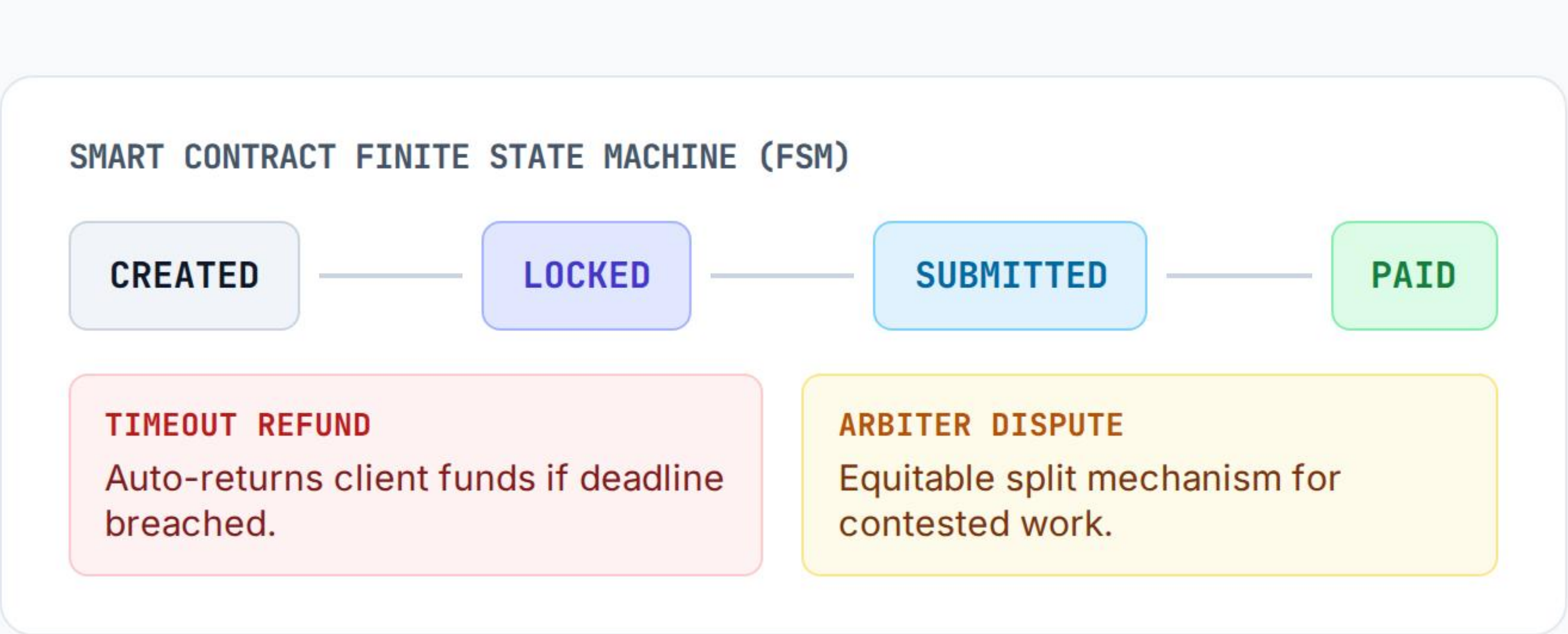
Automatic payout when client review deadline elapses.

100% ghosting immune

Why MicroEscrow Wins Across Every Dimension

CRITERIA	WEB2 (UPWORK/FIVERR)	LEGACY WEB3 ESCROWS	 MICROESCROW
Platform Fee	10% – 20% cut	1% – 3% cut	0% (Pure P2P)
Freelancer Gas Cost	N/A (Strict KYC/Banks)	High (Requires ETH)	\$0.00 Gasless Submissions
Ghosting Defense	Slow manual support	Manual claim required	Automated Smart Release
Settlement Speed	5 – 14 days holding	5 – 15 minutes	< 2 Seconds (Base L2)
Code Transparency	Proprietary black box	Often unverified	100% BaseScan Verified

State Machine & Engineering Rigor



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PASSING

Foundry Unit Test Suite

100% Core Logic and edge-case branch coverage.

Solidity ^0.8.24 + OpenZeppelin v5

Battle-tested ReentrancyGuard and Ownable access.

Live On-Chain Deployment & Verification

BASE SEPOLIA TESTNET (84532)

VERIFIED

Smart Contract Address

0x57099f3125faF6591f23000E8985E18c0c2202Ac

Deployer / Arbiter Address

0xa1bab221F6bFB93AFa3367D8aAA2c7DD5049EC9a

Deployment Transaction Hash

0xb3976987fe4eee2eb5e2d3565cddc78826b16de9a...

Next.js 14 Frontend Stack

App Router, Tailwind CSS, TypeScript, and modern Wagmi v2 + Viem hooks.

Base L2 Sub-Cent Fees

Gas overhead under \$0.01 per milestone lock, making micro-gigs economically viable.

The Ultimate "Labor-to-Crypto" On-Ramp

300 Million

Global University Students

Over 35% already engage in coding, design, and technical freelancing.

Earned Capital, Not Speculation

Students enter Web3 by contributing real engineering labor without upfront token purchases.

Permanent On-Chain Track Record

Each successfully unlocked milestone validates verifiable work history on Base L2.



Strategic Product Roadmap

PHASE 1 (COMPLETED)



Hackathon Core MVP

- Base Sepolia Escrow FSM
- Timeout auto-release protection
- Next.js 14 role-aware UI

PHASE 2 (Q4 2026)



Smart Accounts & SBT

- ERC-4337 Account Abstraction
- Production Paymaster gas sponsors
- Soulbound Token student resume

PHASE 3 (Q1 2027)



Multi-Asset & Court

- USDC & EURC stablecoin locking
- Decentralized jury dispute courts
- Cross-L2 settlement bridging

Empowering Student Builders Without Compromise

MicroEscrow bridges Web2 convenience with Web3 security. Fully verified, audited in Foundry, and deployed live on Base Sepolia.

 **GitHub Repository**

github.com/PHONGUIT22/MicroEscrow

 **BaseScan Smart Contract**

[0x57099f3125faF6591f23000E8985E18c0c2202Ac](https://basescan.org/contract/0x57099f3125faF6591f23000E8985E18c0c2202Ac)

 **Team**

UIT (VNU-HCM)